

Anatomy of Flowering Plants — Revision Notes

Ultra-Condensed High-Yield Revision Sheet for Annual Examination

Author: Narendra Sir — Lecturer in Biology, Founder/CEO BAIO Learning System

1. CORE CONCEPTS & TISSUE SYSTEMS AT A GLANCE

1. Epidermal System

- **Epidermis:** Single outermost layer, parenchymatous, covered by waxy **Cuticle** (absent in roots).
- **Stomata:** Pore + 2 Guard cells (bean-shaped in dicots, dumb-bell in grasses) + Subsidiary cells.
- **Trichomes:** Multicellular stem hairs; check water loss.
- **Root Hairs:** Unicellular epiblema elongations; absorb water/minerals.

2. Ground System

- All tissues except epidermis & vascular bundles.
- Consists of: Parenchyma, Collenchyma, Sclerenchyma.
- **Stems/Roots:** Cortex, Endodermis, Pericycle, Pith, Medullary rays.
- **Leaves:** Photosynthetic ground tissue = **Mesophyll**.

3. Vascular System

- Vascular bundle = Xylem + Phloem.
- **Radial:** Xylem & phloem on different radii (Roots).
- **Conjoint:** On same radius (Stems & Leaves).
- **Open:** Cambium present; secondary growth occurs (Dicot stem).
- **Closed:** Cambium absent; no secondary growth (Monocots).

Key Formulas & Memory Tricks:

- **Stele** = Pericycle + Vascular Bundles + Pith + Medullary rays (all tissues inner to endodermis)
- **Root Xylem = Exarch** (Protoxylem Outside / Centripetal) | **Stem Xylem = Endarch** (Protoxylem Inside / Centrifugal)

2. RAPID HIGH-YIELD COMPARISON TABLES

A. Dicot Root vs. Monocot Root

Feature	Dicot Root (Sunflower)	Monocot Root (Maize)
Xylem Patches	2 to 4 (Diarch to Tetrarch)	Polyarch (More than 6 patches)
Pith (Medulla)	Small, reduced, or inconspicuous	Large and well-developed
Secondary Growth	Present (vascular cambium develops)	Absent (no cambium develops)
Casparian Strips	Present on radial & tangential walls (suberin)	Present on endodermis; passage cells present

B. Dicot Stem vs. Monocot Stem

Feature	Dicot Stem (Sunflower)	Monocot Stem (Maize/Grass)
Hypodermis	Collenchymatous	Sclerenchymatous
Vascular Bundles	Arranged in a ring ; Open , Conjoint	Scattered in ground tissue; Closed , Conjoint
Bundle Sheath	Absent	Surrounded by sclerenchymatous bundle sheath
Water Cavities / Lacunae	Absent	Present (protoxylem water cavity)
Phloem Parenchyma	Present	Absent
Endodermis / Pith	Distinct (Starch Sheath); Pith & rays present	Absent ; Undifferentiated ground tissue

C. Dicot (Dorsiventral) Leaf vs. Monocot (Isobilateral) Leaf

Feature	Dicot Leaf (Dorsiventral)	Monocot Leaf (Isobilateral)
Stomata Location	Hypostomatic (more on abaxial/lower surface)	Amphistomatic (equal on both surfaces)
Mesophyll	Differentiated (Palisade + Spongy parenchyma)	Undifferentiated (uniform parenchyma)
Bulliform Cells	Absent	Present on adaxial epidermis (leaf rolling in grasses)
Venation / Bundles	Reticulate venation; variable bundle sizes	Parallel venation; nearly uniform bundle sizes

3. HIGH-FREQUENCY TERMS FOR 1-MARK & 2-MARK QUESTIONS

Casparian Strips

Deposition of water-impermeable waxy material **suberin** on radial and tangential walls of root endodermis. Acts as physiological barrier controlling apoplastic transport.

Bulliform (Motor) Cells

Large, empty, colourless cells on adaxial leaf epidermis of grasses. When flaccid during water stress, cause leaf rolling to minimize transpiration.

Starch Sheath

The endodermal layer in dicot stems, so named because its cells are characteristically rich in starch grains.

Conjunctive Tissue

Parenchymatous cells located between xylem and phloem patches in roots; participates in cambium ring formation in dicots.

Medullary Rays

Radially placed parenchymatous cell columns between vascular bundles connecting pith with cortex in dicot stems.

Guard Cell Shapes

Bean / Kidney-shaped: In dicots.

Dumb-bell shaped: In monocots (grasses).

4. LAST-MINUTE BOARD EXAM CHECKLIST (4 MARKS MASTER PREP)

- ✓ **3 Differences between Dicot & Monocot Root:** 2–4 xylem vs. Polyarch; Small pith vs. Large pith; Secondary growth present vs. absent.
- ✓ **3 Differences between Dicot & Monocot Stem:** Ring vs. Scattered bundles; Open vs. Closed; Collenchymatous vs. Sclerenchymatous hypodermis.
- ✓ **3 Components of Stomatal Apparatus:** Stomatal pore + Guard cells + Subsidiary cells.
- ✓ **Quick Definitions:** Stele, Exarch vs. Endarch, Radial vs. Conjoint, Casparian strips, Bulliform cells.